SHREEJIT VERMA

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EDUCATION

Georgia Institute of Technology (Online), Atlanta, USA — Aug 2024 – Dec 2026 (exp.)
Master of Science in Computer Science with Specialization in Computing Systems
Coursework: Computer Networks, Advanced Operating Systems, Distributed Computing, Database Management Systems

Stevens Institute of Technology, Hoboken, NJ — Aug 2024 – May 2026
Master of Science in Financial Engineering | **GPA**: 3.974/4.0
Coursework: Market Microstructure, Portfolio Theory and Applications, Algorithmic Trading Strategies, Multivariate Statistics

WorldQuant University, New Orleans, USA — Dec 2021 – May 2024
Master of Science in Financial Engineering | **GPA**: 86%
Coursework: Deep Learning for Finance, Financial Econometrics, Fixed Income, Equity, Portfolio Management, Risk Management

Carnegie Mellon University, Tepper School of Business, New York, USA — Aug 2021 – Oct 2021
Master of Science in Computational Finance – (Program withdrawn due to father's illness)
Coursework: Investments, Statistical Machine Learning, Simulation Methods, Financial Computing, Algorithmic Optimization

Vellore Institute of Technology, Vellore, India
Bachelor of Technology in Computer Science and Engineering | **GPA**: 8.78/10.0 — July 2014 – Sept 2018
Coursework: Data Structures and Algorithms, Programming Language Translators, Natural Language Processing (NLP)

PROFESSIONAL EXPERIENCE

BNP Paribas CIB, New York, USA
C++ Quantitative Developer (Co-op), Automated Market Making — Feb 2026 – May 2026
- Built low-latency components of the automated market-making stack for the Prime Credit Market (average **$500M** of daily market-making volume), spanning real-time market-data ingestion, tick analytics, and pricing/execution paths
- Profiled and optimized the software hot path feeding FPGA-accelerated market-data handlers and quoting engines
- Integrated secure on-premise LLM tooling with Git/Jira/Confluence to automate code, testing, and documentation workflows

LogiNext Solutions Inc., Mumbai, India
Senior Software Engineer, Analytics — Mar 2023 – July 2024
- Architected and developed Map Construction Algorithms, Map Routing Algorithms, and solved Rich Vehicle Routing Problems (3 Nested NP-Hard Problems) using CP-SAT Constraint Programming and Convex optimization over PostGIS, MongoDB and S3
- Led a 12-engineer team delivering a high-throughput geospatial mapping application platform
- Built an LLM-powered debugging and query-resolution tool across the company, cutting mean bug-resolution time by **80%**

Versor Investments (QR Systems LLP), Mumbai, India
Quantitative Developer, Merger Arbitrage and Stock Selection Portfolio — Feb 2022 – Oct 2022
- Developed and backtested systematic merger-arbitrage strategies of an **$8.5 Billion** AUM fund, improving alpha capture by **15%**
- Built and deployed ML pipelines for Order and Execution Management Systems, increasing trade execution efficiency by **29%**
- Designed an ESG-driven merger-arbitrage signal capitalizing on arbitrage opportunities on pre- and post-merger statistics

Bank of America (BA Conitnuum India Pvt Ltd.), Chennai, India
Senior Software Engineer, Fixed Income Commodities and Currencies (FICC) — Jan 2020 – July 2021
- Engineered Python-based trading services enhancing the storage, processing, matching, and execution of trades on QUARTZ
- Integrated C++ pipelines to store trades in the object-oriented database SANDRA, reducing trade processing latency by **50%**
- Led the migration of **1 million** plus lines of code to Python 3.8, enhancing system scalability and execution efficiency by **40%**

Senior Tech Associate in Data Analysis and Insight Technology — June 2018 – Dec 2019
- Architected and developed an ML/AI platform to deploy predictive models, increasing decision-making accuracy by **67%**
- Designed machine learning models for data validation rules prediction, reducing by close to **36 Full-Time Equivalents (FTE**

SKILLS

Mathematics/Statistics: Probability, Stochastic Calculus, Differential Equations, PDE, Linear Algebra, Numerical Methods, Markov
Quantitative Finance: Statistical Analysis, Derivative Pricing, Time Series Analysis, Factor Modeling, Predictive Modeling, Greeks
Machine Learning: Linear Regression, Clustering, Random Forest, XGBoost, RNN, LSTM, Deep Learning, Neural Networks, NLP
Programming: C++ (17/20/23, primary), Python, C, Java, R, MATLAB, JavaScript, Node.js, ReactJS, NumPy, Pandas, Polars, SciPy, Keras, PyTorch, TensorFlow, Scikit-learn, QuantLib, Statsmodels, QtPy, CVXPY, OpenMP, MPI, CUDA, Bash Scripting
Data Engineering: Airflow, Dask, Spark, PySpark, FastAPI, REST APIs, Kafka, Flink, NoSQL, GraphQL, SQL, BQL, KDB+, Q, VBA, ETL, Big Data, Hadoop, HDFS, PostgreSQL, MongoDB, ZeroMQ, Cassandra, Django, Redis, JIRA, Confluence, Agile, Scrum
Computer Science: TCP/IP, UDP, Multicast, Cache and Multithreading optimization, FPGA (Verilog, VHDL), Blockchain
Cloud Computing and DevOps: Linux, Git, Jenkins, CI/CD, Ansible, Docker, Kubernetes, Helm, AWS, GCP, WSL

RESEARCH & SELECTED PROJECTS

AI-Integrated FPGA for Market Making in Volatile Environments (MS Thesis, Stevens) | GitHub — Oct 2025 – May 2026
- Engineered a sub-10µs low-latency trading system, with a custom-built limit order book, FPGA market data handlers, kernel bypass (DPDK), hardware timestamping, and lock-free data structures for deterministic, microsecond-level execution

Sovereign AI Command Center | GitHub — May 2026 – July 2026
- Architected a local-first, autonomous agentic AI platform orchestrating LLM providers behind a unified API, enabling agents to autonomously execute multi-step tool-calling workflows via the Model Context Protocol (MCP) with zero data leaving the host.

- Developed a Retrieval-Augmented Generation (RAG) and persistent semantic-memory system on ChromaDB using high-dimensional vector embeddings for low-latency retrieval, plus a hardware-aware layer that deploys quantized open-weight models

Plug-and-Play Agentic AI Engineering Harness | GitHub May 2026 – July 2026

- Architected a cross-platform agentic AI developer platform using NixOS, declarative configuration, version-pinned dependencies, symlink-managed files, and automated health checks to bootstrap a complete local AI engineering environment on clean machines
- Integrated multi-agent orchestration and agent-native CLI workflows to automate isolated Git worktrees, autonomous task execution, CI-gated shipping (code review/testing/linting, PR in one command), overnight runs, and upstream synchronization

Adaptive Volatility Regime-Based Execution and Risk Framework | GitHub Sept 2025 – Dec 2025

- Developed adaptive volatility regime switch framework dynamically selecting among passive, TWAP, and aggressive strategies
- Achieved **20.0%** increase in Sharpe Ratio, **6.1%** transaction cost reduction, **20.1%** CVaR decrease with robust risk management

Statistical Arbitrage Reversal and Momentum Strategies (Quant Researcher, WallStreetQuants) | GitHub Jun 2025 – Aug 2025

- Designed and backtested a 120-day volume-momentum-based crypto portfolio strategy, yielding a **155.76%** annualized return and a **1.94** Sharpe Ratio (post transaction costs), significantly outperforming the Bitcoin buy-and-hold benchmark

Dynamic Portfolio Optimization (Master's Thesis, MScFE, WorldQuant University) | GitHub Mar 2024 – June 2024

- Built a real-time portfolio optimization system using convex and non-convex optimization methods, enhancing risk-adjusted returns via adaptive asset rebalancing and multi-factor modeling, managing interest rate, FX, credit, and market risks

Environmental Social Governance (ESG) Merger Arbitrage Strategy Apr 2022 – June 2022

- Developed the ESG Strategy, which was subsequently converted into a portfolio and embedded across all existing portfolios. It caters to the arbitrage opportunity being made by the effect of ESG scores on target and acquirer pre- and post-merger statistics

Blockchain In Retail Jan 2018 – Mar 2018

- Developed a decentralized e-commerce platform to secure and streamline retail transactions with smart contracts, currency conversion, custom hashing, and matching algorithms. Using Homebrew, Truffle, MetaMask, Ganache, GETH, Solc, Puppeth

QS Rank Predictor June 2017 – July 2017

- Constructed an ensemble machine learning model consisting of multiple Deep neural networks to predict QS World Ranking
- The model also gave suggestions on areas to improve, helping in achieving a world ranking of 301 - 400 for the VIT CS in 2020

ACHIEVEMENTS

1st Place for personal portfolio in Vanguard ETF Trading Challenge | **President** of Stevens Graduate Financial Association | State Rank Holder in **International Science Olympiad** and **International Mathematics Olympiad | Beta Gamma Sigma** Member

Global Recognition Gold Award (Bank of America)

- Led enterprise-wide AI/ML campaign identifying 64 high-impact use cases on fraud detection, forecasting, and optimization
- Organized 4 large-scale Data Science events across Bank of America, delivering AI/ML lectures to 2500+ employees

Global Recognition Silver Award (Bank of America, 2x)

- For contribution in Total Return Swap Bonds, Futures, Options, and Cash (LOBs) under the Post Trade Processing Team
- Architected and developed AI/ML framework integrating end-to-end tools for building in-house Data Science Projects

CERTIFICATIONS

*Links to certificates are attached as hyperlinks to the respective course names

- **Finance:** CFA Level 1; Bloomberg Market Certification; Financial Engineering and Risk Management Part I & II) (Coursera); Investment Foundations Program (CFA, USA); **The Complete Financial Analyst Training & Investing Course** (Udemy); **Machine Learning for Trading Specialization** (Google Cloud Platform, New York Institute of Finance): (Introduction to Trading, Machine Learning & GCP, Using Machine Learning in Trading and Finance, Reinforcement Learning for Trading Strategies); **Investment Management Specialization** (University of Geneva, UBS): (Understanding Financial Markets, Meeting Investors' Goals, Portfolio and Risk Management, Securing Investment Returns in the Long Run, Planning your Client's Wealth over a 5-year Horizon); **Trading Strategies in Emerging Markets Specialization (**Indian School of Business, ISB): (Trading Basics, Trading Algorithms, Advanced Trading Algorithms, Creating a Portfolio, Design your own trading strategy – Culminating Project); **Finance & Quantitative Modeling for Analysts Specialization** (University of Pennsylvania, Wharton): (Fundamentals of Quantitative Modeling, Introduction to Spreadsheets and Models, Financial Acumen for Non-Financial Managers, Introduction to Corporate Finance); **Corporate Finance and Valuation** (NYU STERN, Aswath Damodaran)
- **Computer Science: Deep Learning Specialization**: (Statistical Inference, Regression Models, Practical Machine Learning, Developing Data Products, Data Science Capstone); **Applied Data Science with Python Specialization** (University of Michigan): (Introduction to Data Science in Python, Applied Plotting, Charting & Data Representation in Python, Applied Machine Learning in Python, Applied Text Mining in Python, Applied Social Network Analysis in Python); **Data Science Foundations using R Specialization** (Johns Hopkins University): (The Data Scientist's Toolbox, R Programming, Getting and Cleaning Data, Exploratory Data Analysis, Reproducible Research); **Data Science Statistics and Machine Learning Specialization:** (Statistical Inference, Regression Models, Practical Machine Learning, Developing Data Products, Data Science Capstone); **Big Data Specialization** (University of California San Diego): (Introduction to Big Data, Big Data Modeling and Management Systems, Big Data Integration and Processing, Machine Learning with Big Data, Graph Analytics for Big Data, Big Data - Capstone Project); **Data Structures and Algorithms Specialization**: (Algorithmic Toolbox, Data Structures, Algorithms on Graphs, Algorithms on Strings, Advanced Algorithms and Complexity, Genome Assembly Programming Challenge); **Algorithms, Part I** & **Part II** (Princeton University)

Interests: Chess, Poker, F1, Martial Arts, Cricket, Boxing, Badminton, Reading, Cooking, Dancing, Psychology, History, Philosophy